Local certificates for Diophantine consistency

Shared Pell orientation and smaller quartics with unit coefficients

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Abstract

We give explicit ordinary integer polynomials whose lack of integer solutions is provably equivalent in ZFC to the syntactic consistency of ZFC. The work builds on the direct numerical-calculus branch of ProveIt and develops two reductions. First, a linear trace-orientation term is added to an existing pooled Pell-index congruence. This shares the work formerly performed by nine separate sign certificates. Second, coefficient-preserving identifications of graph aliases are certified by exact polynomial collection; a global four-cycle restriction on an auxiliary factor graph is unnecessary. A two-variable quartic equality energy removes further auxiliary variables. The release contains the literal residuals and complete collected coefficient lists, constructive zero-set arguments, deterministic reconstruction code, and independent finite checkers. Every final variable ranges over all of $\mathbb{Z}$. The reductions are ordinary mathematical proofs formalizable in ZFC, composed with the explicitly identified inherited C222/ZF/ZFC calibration. They do not claim a new end-to-end proof-assistant verification or a globally simplest equation.

1 The actual equations and the scope of the result

The principal direct equation is the fixed polynomial

$$P_{\mathrm{dir}}(z_0,\dots,z_{305}) = \sum_{j=0}^{205} r_j(z)^2 = 0, \qquad z \in \mathbb{Z}^{306}. \tag{1}$$

Here the r_j are all the quadratic residuals in `direct/integer306/equation.json`, in their specified coordinate order. Their complete readable transcription is supplied with the release. Its exact degree is four, its collected support has 6,259 monomials, and its coefficient height is 128,446. This is a finite literal equation: there is no variable-exponent operation, binomial predicate, unspecified universal polynomial, or additional implicit existential variable in (1).

The unit-coefficient equations have the particularly uniform form

$$P_{\mathrm{unit}}(x) = 1 + \sum_{\alpha\in\mathcal{A}_+} x^\alpha - \sum_{\alpha\in\mathcal{A}_-} x^\alpha = 0, \qquad x \in \mathbb{Z}^n. \tag{2}$$

The two finite, disjoint sets of nonconstant exponent vectors are exactly the positive and negative entries of each released `polynomial.json`. Repeated coordinate indices mean powers; the empty index tuple denotes the constant. The coefficient list itself, not a request to generate some unspecified polynomial, defines the equation. The selected profiles and their exact metrics are listed below.

For every selected polynomial P of arity n, our conclusion is

$$\mathsf{ZFC} \;\vdash\; \big[\neg\exists x \in \mathbb{Z}^n\; P(x) = 0\big] \;\longleftrightarrow\; \mathrm{Con}(\mathsf{ZFC}). \tag{3}$$

Construction	Variables	Degree	Terms	Height
Parent direct quartic [131]	324	4	5,808	128,446
New direct quartic	306	4	6,259	128,446
Parent unit-coefficient quartic [127]	2,215	4	30,468	1
New graph, same 442-coordinate source	1,252	4	27,738	1
New graph, fewer variables	985	4	95,609	1
New graph, smaller support	1,122	4	25,922	1

Table 1: All variables are unrestricted integers. Terms are fully collected nonzero monomials. The graph companions permit individual fourth powers.

The proof proceeds by proving the positive existential statement equivalent to the existence of a finite syntactic ZFC refutation and then negating it. No assertion of either side of (3) is needed.

1.1 What "simpler" means here

There is no total ordering of all the relevant costs. We report arity, degree, number of collected monomials, and coefficient height separately. In particular, the repository already has quartics with fewer variables and very large fixed coefficient tuples, and higher-degree equations with far fewer variables [3]. The direct equation improves the small-coefficient C222 construction; the graph equations improve the unit-coefficient quartic construction. None is asserted to minimize arity among all consistency polynomials.

For a nonzero polynomial with integer coefficients, coefficient height cannot be less than one. Thus the graph equations have the smallest possible nonzero coefficient height, while their arity and support remain subjects for optimization. This elementary height observation is quite different from a global simplicity theorem.

The source snapshot is `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. The repository's main catalogue at that snapshot still summarizes earlier waves, although packages 127–133 have already been placed. We inspected those newer packages directly. Comparisons are to specified literal artifacts, so a change of source, domain, or individual-degree bound is stated explicitly.

2 An elementary graph representation of equations

We first describe the inherited graph representation, because it exposes exactly what the new contraction theorem preserves. Let $\mathcal{G} = (V, E)$ be a finite simple graph, with an arbitrary orientation. Attach an integer polynomial m_e of degree at most two to each edge. Put

$$d_v = \sum_{e \text{ leaves } v} m_e - \sum_{e \text{ enters } v} m_e, \qquad \mathcal{E}_{\mathcal{G}} = \frac{1}{2} \sum_{v \in V} d_v^2.$$

The expression is nonnegative over $\mathbb{R}$ and vanishes precisely when every vertex divergence vanishes. The factor $1/2$ is proof notation: after collection the supplied final polynomial has integral coefficients.

Before substituting the edge polynomials, consider independent edge symbols t_e. Each t_e^2 occurs at its two endpoints and therefore has coefficient one in $\mathcal{E}_{\mathcal{G}}(t)$. Two distinct edges in a simple graph share at most one endpoint, so their product has coefficient $+1$, -1, or zero. The resulting height-one assertion need not survive an arbitrary substitution $t_e = m_e$; different edge products may become the same monomial. That is the only coefficient issue the new exact collector must settle.

Lemma 2.1 (Integral tree flows)**.** *Partition the vertices into connected blocks whose internal edges form trees. Give all edges between blocks fixed integer values. There is an integer assignment to the internal edges making every divergence zero if and only if each block has total external signed flow zero. When it exists, the assignment is unique.*

Proof. Summing divergences over a block cancels all its internal edges, proving necessity. For sufficiency, root its tree and remove leaves successively. The only remaining internal edge at a leaf has coefficient $+1$ or -1, so the leaf equation uniquely determines its integer value. Transfer this contribution to the parent and continue. The final root equation is the assumed block balance. The construction proves integrality and uniqueness. □

Normalize a source system into rows

$$R_r = \sum_{i=1}^{s_r} \epsilon_{ri} m_{ri}, \qquad \epsilon_{ri} \in \{-1, 1\}, \quad \deg m_{ri} \le 2.$$

The normalization uses monic topological equations to construct positive integer coefficients by binary addition and to construct constant multiples of source variables. The source coordinates are retained, every added coordinate has a unique integer value, and the normalization therefore preserves the integer zero set bijectively.

For a chosen positive integer H, use one hub block of H vertices. The block of row r has $\lceil s_r/H \rceil$ vertices. Represent each signed term occurrence by a distinct edge between its row block and the hub block; the available bipartite capacity is sufficient. Connect each block internally by a tree. Row-block balance is exactly $R_r = 0$, and hub-block balance is the negative sum of those equations. Lemma 2.1 gives the desired equivalence. The number of independent tree-flow coordinates is

$$H - 1 + \sum_r (\lceil s_r/H \rceil - 1). \tag{4}$$

Each normalized logical variable is represented on its occurrences by one or more *aliases*. An alias is an ordinary independent integer coordinate until its equality energy is imposed. Terms are homogenized to degree two using aliases of the constant one. Tree-flow variables occupy a separate coordinate class. There is no hidden interpretation imposed on the quantified coordinates: all alias and unit conditions are equations inside the final polynomial.

3 A smaller equality certificate

For $k \ge 3$ aliases of one logical coordinate, use the cycle energy

$$C(a_1, \dots, a_k) = \sum_{i=1}^{k} a_i^2 - \sum_{i=1}^{k} a_i a_{i+1}, \qquad a_{k+1} = a_1. \tag{5}$$

Twice this polynomial is $\sum_i (a_i - a_{i+1})^2$. It is nonnegative and vanishes exactly when all aliases agree. Its collected nonzero coefficients are $+1$ or -1. A singleton needs no equality equation. A two-vertex cycle would have coefficient -2 on $a_1 a_2$, so the earlier compiler inserted a third, dummy alias.

Lemma 3.1 (Two-coordinate quartic equality)**.** *The polynomial*

$$F(a, b) = a^4 + b^4 - a^3 b - a b^3 \tag{6}$$

is nonnegative over $\mathbb{R}$ and vanishes exactly when $a = b$. Its four nonzero coefficients have absolute value one. In particular it requires no dummy coordinate.

Proof. Two useful exact identities are

$$F(a,b) = (a-b)^2(a^2+ab+b^2),$$
$$2F(a,b) = (a^2-b^2)^2 + (a^2-ab)^2 + (ab-b^2)^2.$$

The first proves the equality condition because $a^2+ab+b^2 = (a+b/2)^2 + 3b^2/4$ is positive except at $(0,0)$, where $a=b$ already holds. The second gives a direct rational sum-of-squares certificate and is used by the independent collector. □

The new equality energy must be checked against the other quartic monomials. For example, if a^4 already has coefficient $+1$ in the graph energy, adding (6) would produce coefficient $+2$. Such a replacement is rejected. The compiler applies the lemma only where the complete collected polynomial keeps height one; otherwise it retains the three-alias quadratic cycle. This is an exact finite condition, not an assumption that the new monomials occupy a fresh namespace.

For unit aliases $u_1, \dots, u_k$, with $k \geq 2$, use

$$U = 1 + \sum_{i=1}^{k} u_i^2 - u_1 - u_k - \sum_{i=1}^{k-1} u_i u_{i+1}. \tag{7}$$

Twice U is $(u_1-1)^2 + \sum_{i<k}(u_i - u_{i+1})^2 + (u_k-1)^2$. It is nonnegative, and its zeros make every unit alias equal to one.

4 Exact coefficient collection replaces a global restriction

The earlier factor-sharing compiler keeps a bipartite factor graph without four-cycles [1]. That condition ensures that every product of two source-edge monomials determines its two occurrences uniquely. It is a convenient sufficient condition, but it also restricts products which never appear in the graph energy.

For instance, take four pairwise disjoint source edges with monomials as, at, bs, bt. Their factor graph contains a four-cycle, yet their actual energy is

$$(as)^2 + (at)^2 + (bs)^2 + (bt)^2.$$

All coefficients are one. The algebraic equality $(as)(bt) = (at)(bs)$ is irrelevant because neither cross product occurs. More generally, even colliding contributions can be harmless when their signed collected coefficient remains in $\{-1, 0, 1\}$.

Definition 4.1 (Admissible alias identification)**.** An identification maps old aliases to new aliases only within the same logical source-variable class, or within the unit class. Flow coordinates are retained separately. Substitute this map into the graph-edge polynomials, reconstruct the equality cycles and unit path, and optionally use Lemma 3.1 for two-alias classes. The identification is *admissible* if exact expansion shows that every collected coefficient of the complete resulting energy is 0, 1, or -1.

Theorem 4.2 (Certified alias quotient)**.** *Let a normalized integer source system have a certified graph representation as above. Every admissible alias identification produces a polynomial of degree at most four and coefficient height at most one, whose integer zeros are in bijection with the integer zeros of the normalized source. If its collected list contains a degree-four monomial and constant coefficient one, its degree and coefficient height are exactly four and one.*

Proof. Write the resulting polynomial as

$$P' = \frac{1}{2}\sum_{v} d_v'^2 + \sum_{\text{cycle classes}} C + \sum_{\text{pair classes}} F + U.$$

Every summand is nonnegative over $\mathbb{R}$. Thus $P' = 0$ forces every divergence, every alias equality, and every unit condition. A new alias class therefore has one well-defined logical source value. Substitute these values into the source term occurrences. Their logical identities are unchanged by the identification. Summing the zero divergences over each row block recovers all normalized source equations.

Conversely, take an integer zero of the normalized source. Assign each new alias its logical source value and each unit alias the value one. Every equality and unit energy is zero. The same external row-term values satisfy all block balances, so Lemma 2.1 supplies the unique integer internal flows. Consequently $P' = 0$.

Every alias and internal flow is uniquely determined by the source zero, including any retained dummy aliases. Reading source values and performing this construction are inverse maps. This proves bijection. The degree bound follows from the degrees of the graph edges and equality energies. The coefficient assertion is precisely the exact finite admissibility check. A literal degree-four coefficient and the constant one witness attainment of both bounds. □

Remark 4.3 (Zero sets and polynomial values). The theorem does not claim the off-solution identity $P'(L(z)) = \sum_r R_r(z)^2$. Canonical rooted-tree elimination leaves one residual per row block and $-\sum_r R_r$ at the hub root; its graph energy is instead

$$\tfrac{1}{2}\left(\sum_r R_r^2 + (\sum_r R_r)^2\right).$$

Both have the same zero set. Distinguishing these identities prevents an incorrect claim of value-preserving substitution.

4.1 How the particular certificates were found

The search merges two aliases of the same logical value and updates all affected collected monomials exactly. A proposed merge is accepted only if all new coefficients stay in $\{-1, 0, 1\}$. Equality cycles are rebuilt on the distinct surviving aliases. Each safe two-coordinate quartic replacement is then checked against the full polynomial. All computation uses integers, with no floating-point comparisons or random evaluations as proof substitutes.

The order of attempted merges is an optimization choice. A different order can give a different valid quotient; neither the deterministic order nor the finite hub sweep proves optimality. The accepted certificate is checked independently of the search history. This is essential: trust in a greedy search is unnecessary once its exact finite output has been verified.

The parent graph, the map of source occurrences, the parent-to-child alias map, all new equality classes, every edge, and the final coefficient list are retained. The independent checker expands twice the energy as a sum of squares and compares every coefficient. Its expansion does not use the producer's adjacent-edge collection routine.

5 A direct quartic in 306 unrestricted integers

The main direct construction starts from the literal centered polynomial of package 131 [2], in 324 unrestricted integer coordinates. The source is pinned to ProveIt commit `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`; its equation, centering certificate, and ordinary arithmetic proof are retained in the archive. The new polynomial is

$$P_{306}(z_0,\dots,z_{305}) = \sum_{j=0}^{205} r_j(z)^2, \qquad z \in \mathbb{Z}^{306}. \tag{8}$$

Every r_j is an explicitly listed integer polynomial of degree at most two. There are no implicit domain predicates: all 306 coordinates are unrestricted. The literal residuals and the complete collected coefficient list are supplied, together with their independent verifier.

Equation	Variables	Rows	Degree	Terms	Height
Pinned parent	324	208	4	5808	128446
New equation	306	206	4	6259	128446

Thus 18 coordinates and two rows are removed at the cost of 451 collected terms; both coefficient height and residual height, respectively 128446 and 256, are preserved. The comparison is within this direct C222 construction. Much smaller variable counts exist in the repository with different degrees and coefficient architectures.

Theorem 5.1. *Let P_{324} be the pinned parent polynomial. Ordinary mathematics formalizable in ZFC proves*

$$(\exists z \in \mathbb{Z}^{306}\ P_{306}(z) = 0) \quad \Longleftrightarrow \quad (\exists w \in \mathbb{Z}^{324}\ P_{324}(w) = 0).$$

Consequently, composing with the retained parent calibration,

$$\mathrm{ZFC} \vdash \left[\neg\exists z \in \mathbb{Z}^{306}\ P_{306}(z) = 0\right] \longleftrightarrow \mathrm{Con(ZFC)}.$$

The new idea is a linear orientation term inside the existing bounded Pell-index congruence. It permits nine sign graphs to be replaced by one strict product. The proof must recover the signs before invoking the parent's complete arithmetic theorem. We give that order explicitly.

5.1 Linear orientation in a bounded residue pool

For $A \geq 2$ let $X_j(A), Y_j(A)$ be the canonical Pell sequences:

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2-1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

The unchanged parent square frame and complete anchor yield, independently of the nine encoded-coordinate signs,

$$W \geq 258, \qquad A > W^W, \qquad A - 1 > 4W^3. \tag{9}$$

They also give 15 principal Pell norms, absolute indices $0 \leq j_i < W$, signed proposed indices $|k_i| < W$, and absolute bounds $|n|, |m| < W$ for every requested base and output. The fully indexed anchor is at W; its unchanged cap bounds every principal ordinate in absolute value. Its proof uses the frame, the anchor common-multiplier graph and the cap graph, all retained here. No $k_i \geq 0$ premise is required.

Pell classification supplies signs $\sigma_i, \epsilon_i \in \{1, -1\}$ with

$$x_i = \sigma_i X_{j_i}(A), \qquad y_i = \epsilon_i Y_{j_i}(A).$$

At $j_i = 0$ the choice of ϵ_i is immaterial. Modulo $A - 1$, the recurrence gives $X_j \equiv 1$ and $Y_j \equiv j$.

Lemma 5.2 (Linear orientation pool)**.** *Assume the preceding bounds. Let I be a set of nine of the fifteen principal pairs. The single row*

$$\sum_{i=1}^{15}(y_i-k_i)^2+\sum_{i\in I}(1-x_i)=(A-1)q_0,\qquad q_0\in\mathbb{Z},\tag{10}$$

forces $\epsilon_i j_i=k_i$ for every i and $\sigma_i=1$ for $i\in I$.

Proof. Reduce (10) modulo $A-1$. It makes $A-1$ divide

$$E=\sum_{i=1}^{15}(\epsilon_i j_i-k_i)^2+\sum_{i\in I}(1-\sigma_i).$$

Every term in the last sum belongs to $\{0,2\}$. Therefore

$$0\le E\le 60(W-1)^2+18<4W^3<A-1.$$

For example, $4W^3-60(W-1)^2-18=4W^2(W-15)+120W-78>0$ for $W\ge 258$. Hence $E=0$, and every nonnegative summand vanishes. □

Only the residue representative E is small and nonnegative. The actual left side of (10) need not be either. In particular the pool quotient must remain unrestricted. A squared term $(x_i-1)^2$ was used in the parent's conservative orientation variant; the linear term $1-x_i$ works because its residue has the nonnegative alphabet $\{0,2\}$. The new application combines this cheaper orientation with removal of the nine encoded-coordinate sign graphs.

5.2 Signed indices and the early bootstrap

We need a power lemma that treats zero and negative proposed indices. Write $Y_{-j}=-Y_j$ and $X_{-j}=X_j$.

Lemma 5.3 (Signed-index power request)**.** *Let $2\le n<W$, $|k|<W$, $|m|<W$, and put $T=2An-n^2-1$. Suppose*

$$y=Y_k(A),\qquad x=\pm X_{|k|}(A),\qquad x+(n-A)y=m+Tz$$

for an integer z. If $x=+X_{|k|}$, then $k\ge 0$ and $m=n^k$. If $k>0$, either choice initially allowed for x is forced to be positive. At $k=0$ and positive x, the output is exactly $m=1$.

Proof. For $j\ge 0$ put $H_j=X_j+(n-A)Y_j$ and $J_j=X_j-(n-A)Y_j$. Direct multiplication and the recurrence give

$$H_jJ_j=1+TY_j^2,\qquad H_{j+1}=nH_j+TY_j,$$

so $H_j\equiv n^j \pmod T$. The frame bounds imply $T>W^W$. For $1\le j<W$ the required differences have absolute value below W^W; in particular $|mn^j-1|\le(W-1)^W+1<W^W$.

If $k=-j<0$ and $x=+X_j$, the request gives $mn^j\equiv 1 \pmod T$, hence $mn^j=1$ as an integer equality. This is impossible since $n^j\ge 2$. If $k=j>0$ and $x=-X_j$, it similarly gives $mn^j=-1$, also impossible. The remaining positive-index branch gives $m\equiv n^k \pmod T$, and the size bound makes it equality. At $k=0$, $x=1, y=0$ gives $m=1$ directly by the same small-residue argument. □

The selected nine pairs are precisely the requests at base $q+1$ for the eight encoded CRT columns and their reciprocal-index column. The following bootstrap does not use any of their removed sign graphs. The retained rows and retained nonnegative quantities r,s,o give

$$t=16+h_0^2\ge 16,\qquad B=r+s+1\ge 1,\qquad q=B+o+1\ge 2.$$

The t-pair has its signed intended index from Lemma 5.2. Its genuine request at base q and positive index t forces its abscissa positive by Lemma 5.3. The same pair therefore gives

$$V_0 = (F^2+1)^t,$$

including the possible base-one case before the sign of F is known. For that case the same recurrence gives $H_t \equiv 1 \pmod{2(A-1)}$; the bound $|V_0| < W$ then forces $V_0 = 1$. The literal ceiling rows give $FB = V_0 + s > 0$, so $F \geq 1$. With $b = F^2+1$, both $b > F$ and $b^{t-1} > F$, whence

$$FB \geq b^t > Fb, \qquad B > F^2+1. \tag{11}$$

The unchanged cofinal row $y = V_0(V_0-1)$ also gives $y > F$; the x-sign graph remains present. Neither factorial meaning nor CRT interpretation has been used at this stage.

Lemma 5.2 orients the nine selected abscissas, and Lemma 5.3, at base $q+1 \geq 3$, gives

$$k_a, k_b, k_c, k_d, k_S, k_{S'}, k_u, k_{v_q}, k_{\mathrm{index}} \geq 0.$$

Zero remains a genuine possibility at this point. The reciprocal-index rows give

$$B\,v = F^2 + k_{\mathrm{index}} + 1$$

for an unrestricted integer v. If $k_{\mathrm{index}} = 0$, its right side lies strictly between zero and B by (11), an impossible integer multiple of B. Therefore $k_{\mathrm{index}} \geq 1$.

5.3 Six products and one fused positivity row

The other eight indices are nonnegative. Introduce six unrestricted coordinates $P_2, \ldots, P_7$ with

$$\begin{aligned} P_2 &= k_a k_b, & P_3 &= P_2 k_c, & P_4 &= P_3 k_d, \\ P_5 &= P_4 k_S, & P_6 &= P_5 k_{S'}, & P_7 &= P_6 k_u. \end{aligned}$$

Finally introduce three unrestricted integers a, b, c and impose

$$P_7 k_{v_q} - 1 = a^2 + b^2 + c(c+1). \tag{12}$$

The right side is nonnegative at every integer triple. The product of the eight nonnegative indices is thus at least one, so every index is at least one. This restores all nine old encoded-coordinate signs. Every displayed row is quadratic; a separate P_8 coordinate is avoided.

Conversely every nonnegative integer n is $a^2 + b^2 + c(c+1)$. Indeed, $4n+1$ is not of the forbidden form $4^u(8v+7)$, so Legendre's three-square theorem represents it by three squares. Exactly one is odd; writing the three terms as $(2a)^2, (2b)^2, (2c+1)^2$ gives the result. Thus every old zero, whose eight indices are positive, has witnesses for (12).

The eight-factor idea has a predecessor in the repository's `signed-pell-cap-refined-scalar-product` proof. That proof obtains nonnegative indices from a different residue-one chart. Here the original raw remainder chart is retained; the linear orientation pool supplies the required signs directly. The change is this complete composition and its literal smaller polynomial, not invention of positive-product reasoning.

5.4 Literal chart and exact witness transformations

The centered parent has the following requested-index chart. The third column lists its nine selected abscissa coordinates; all have zero centering shift.

Index	Expression	Abscissa coordinate
k_a	$z_{10} - 1$	174
k_b	z_{11}	180
k_c	z_{12}	184
k_d	z_{13}	188
k_S	z_{14}	192
$k_{S'}$	z_{15}	200
k_u	z_{16}	204
k_{v_q}	z_{17}	212
k_{index}	z_{18}	216

The old nonnegative quantities are $k_i - 1$. Delete precisely parent rows 184–192, the nine scalar graphs labeled `domain_guard.3` through `domain_guard.11`, and their 27 private coordinates 252–278. They occur nowhere else. Modify parent row 180, the pooled-index row, by adding precisely

$$9 - z_{174} - z_{180} - z_{184} - z_{188} - z_{192} - z_{200} - z_{204} - z_{212} - z_{216}.$$

The private pool quotient is coordinate 242 and occurs only in this row. The actual Pell center is $A = z_{151} + 1$, so its divisor is $A - 1 = z_{151}$.

For a new zero, the preceding argument proves the nine missing signs. Let $T_0 = \sum_{i \in I}(1 - X_i)$. Each selected abscissa is canonical, so $T_0/(A-1)$ is an integer. Set

$$q_{\mathrm{old}} = q_{\mathrm{new}} - T_0/(A-1),$$

and choose fresh Legendre triples for all nine deleted scalar graphs. Every other retained parent coordinate stays fixed. The private quotient change restores the old pooled row and affects no other row. We now have a literal zero of the entire parent polynomial, so its complete C222 calibration may be invoked.

For the reverse direction, a parent zero has positive intended indices and canonical abscissas by its ordinary power theorem. Set

$$q_{\mathrm{new}} = q_{\mathrm{old}} + T_0/(A-1),$$

forget the 27 old guard witnesses, calculate the six product coordinates, and supply a triple for (12). All new rows hold. These maps preserve the native front and every retained parent coordinate except the private pool quotient. The Legendre witnesses are not unique, so there is no claim of a bijection between individual full zero tuples. This proves Theorem 5.1.

The parent projection and centering are also part of the supplied certificate boundary. The nine deleted guard rows survived projection; no restoration polynomial depends on their removed auxiliary coordinates or the private pool quotient. The independently checked occurrence sets therefore justify using the parent frame and anchor before restoring those graphs, without a circular appeal to the entire parent theorem.

5.5 Finite certification and its scope

There are $324 - 27 + 6 + 3 = 306$ coordinates and $208 - 9 + 6 + 1 = 206$ residuals. Their sum of squares has a nonzero degree-four coefficient, so the degree is exactly four. The independent verifier imports no

producer code. It checks every retained row, the complete changed pool row, every removed witness occurrence set, the quotient's privacy, the affine chart, the six products, the fused positivity row, all collected coefficients and the exact metrics. It also rejects deliberate changes to an orientation term, a factor shift, quotient privacy, and an expanded coefficient.

These finite checks certify literal algebra and artifact bindings. They do not establish unbounded number theory by sampling. Pell classification, the retained full anchor, Legendre's theorem and the ordinary parent C222/ZF/ZFC calibration remain the identified mathematical dependencies. No new end-to-end proof-assistant certificate or encoded object-ZFC proof is claimed.

6 Logical calibration and what is inherited

The direct source is the ordinary numerical proof-calculus construction called C222 in ProveIt. The relevant logical interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The arithmetic source relates an integer zero of its explicit residual system to a finite C222 derivation. The cited interfaces then relate such a derivation to a finite syntactic ZFC refutation. These are inherited calibration results, with the precise source snapshot and local retained proofs recorded in the provenance ledger.

The new steps in this report are the shared orientation/sign construction, the exact alias-quotient criterion and its realized contractions, and the two-coordinate equality energy as used in that compiler. They use the already supplied source arithmetic calibration. A correctness theorem for a compiler or an exact checksum is not itself a proof of that calibration; our verification reports keep these roles separate.

All new arguments are finite algebra, integer congruences, elementary Pell recurrence arguments, finite graph induction, and explicitly inherited number-theoretic representation theorems. They can be expressed and proved in ZFC. Composing them with the inherited calibration gives

$$\mathsf{ZFC} \vdash (\exists z\ P_{\mathrm{dir}}(z) = 0) \longleftrightarrow (\exists p\ \mathrm{Refutation}_{\mathsf{ZFC}}(p)),$$

and the graph theorem gives the same statement for each unit-coefficient equation. Classical negation yields (3).

There is no newly generated Lean or Rocq proof term for the entire final coefficient file, and no claim that a newly emitted object-language ZFC derivation has been checked by a kernel. The release provides an ordinary proof with exact finite arithmetic certificates. Its mathematical conclusion does not depend on assuming consistency; interpreting the equations as having no integer solution does require the corresponding consistency claim, which we do not prove here.

The direct branch does not simulate K215 and does not instantiate Jones's universal polynomial. Those other branches remain useful comparisons, not requirements of this construction. General universal-pair results such as [5, 4] address other arity/degree tradeoffs and do not imply that the present polynomials are globally minimal.

7 Exact artifacts and verification

The delivery is designed to make each mathematical claim reviewable as a finite object. The original parent residual system is retained without alteration. The direct transformation certificate specifies every deleted guard, changed pooled row, new product equation, and coordinate renaming. The graph certificates retain the complete normalized source and graph interfaces. All expanded polynomials are supplied in sparse integer form.

For the direct equation, the first independent checker verifies 929 literal conditions and rejects four deliberately altered artifacts: a changed orientation term, a wrong first-factor shift, an escaping private quotient, and a changed collected coefficient. A separately written reviewer checks all 199 retained rows, all seven new rows, the nine actual raw power ports, and all 216 parent restoration polynomials. None of those restorations depends on the removed private guard coordinates or the changed private pool quotient. The second reviewer re-expands the full quartic by unordered monomial pairs, independently of the producer's ordered-pair algorithm. The coefficient of z_0^4 is 85, a literal degree-attainment witness.

For each graph profile the checker first verifies the original normalized source and graph certificate. It then checks the complete logical alias map, separate flow coordinates, regenerated equality constraints and unit path. It reconstructs twice the final polynomial as a sum of squares, including the three-square identity for each quartic pair guard, and compares every collected coefficient. It also parses every term of the readable `polynomial.txt` back to the normative coefficient list.

All three graph profiles have been reconstructed from their source rows, with byte-identical parent polynomials, parent certificates, final polynomials, final certificates, and metric files: eighteen mathematical JSON artifacts in total. The optional reconstruction uses the builders only after the independent verification. Normal and optimization-mode Python runs retain all necessary checks; the verifiers do not depend on `assert` statements. The integrated reconstruction also recreates all six direct-equation files in a temporary directory and compares their exact bytes.

The graph evaluator also supplies a canonical witness lift. For an arbitrary integer source tuple z, its construction satisfies the exact identity

$$2P_{\mathrm{unit}}(L(z)) = \sum_r R_r(z)^2 + \left(\sum_r R_r(z)\right)^2 .$$

On source zeros this becomes a constructive lift to graph zeros. The identity follows from the block-tree construction and the monic coefficient normalization; finite evaluations are supplementary diagnostics.

The integrated release check additionally compares the 306-variable source copied into the graph component with the direct equation byte for byte, parses every printed LaTeX residual back into its exact coefficient map, and checks the equation identities and integrity inventory. Its receipt is `verification/RELEASE_CHECK.json`; the optimization-mode companion is `verification/RELEASE_CHECK_OPTIMIZED.json`. Neither check is a search for a zero or a finite test of consistency. The retained build receipts precede final inventory sealing; running the released checker also verifies the completed `MANIFEST.sha256`.

The distinction between a mathematical proof and an integrity check matters. A SHA-256 inventory detects changed bytes; it does not establish an algebraic identity. Exact coefficient reconstruction proves the stated finite identity; it does not prove the unbounded Pell or logical calibration by testing examples. The release includes these layers separately and records which programs check which assertions.

7.1 Offline use

Python 3.10 or later with its standard library suffices for the selected arithmetic-artifact checks. From the release directory run

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --rebuild
python3 -B evaluate.py --help
```

The verification entry point preserves the delivered equations. Evaluation at a supplied integer tuple is exact, but is not an algorithm for deciding whether any integer zero exists. The ordinary proof and

inherited calibration give the semantic interpretation of the polynomial; the evaluator simply evaluates its literal coefficients.

The paper is built with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The LaTeX inputs, generated tables, and readable equation transcriptions are included. PDF byte identity can depend on TeX versions and metadata; the mathematical coefficient lists and their canonical reconstruction are the normative finite artifacts.

8 Further research questions

Joint choice of graph and aliases. The hub layout affects which edge products occur, and hence which alias identifications are safe. The current search chooses a graph first and contracts aliases afterwards. A joint optimizer could trade a small number of tree-flow coordinates for many fewer factor aliases, or reduce support without increasing arity. A graph modification would need a new block-tree certificate, not only a coefficient count.

Larger certified contractions. A greedy sequence of individually admissible pair merges can miss a simultaneous merge whose intermediate polynomials have height two but whose final signed collisions cancel back to height one. Theorem 4.2 applies directly to the final identification. Searching blocks of aliases with an independently checked final certificate is therefore legitimate. There is no need to insist that every hypothetical intermediate quotient itself be released as a height-one polynomial.

Other equality energies. The quartic identity in Lemma 3.1 is useful but collides with fourth powers already present in some graph energies. Can a small library of nonnegative height-one equality polynomials, used with exact overlap tests, eliminate more dummies? Any alternative must preserve its full zero set over unrestricted integers, including zero and negative values. An individual-degree-two companion would be especially useful, since the present quartic pair guard permits fourth powers.

A general finite-residue orientation compiler. The Pell trace has the tiny residue alphabet $\{-1, 1\}$ modulo $A - 1$. The new linear defect $1 - x$ represents the nonnegative values 0 and 2 on that alphabet. Other arithmetic modules may have similarly small residue alphabets, permitting their branch choices to be certified in one pooled congruence. The required error bound and dependency order must be proved before any sign information is used.

Formal certificates for the entire chain. The graph and alias theorems are natural targets for Lean or Rocq. A small verified sparse-polynomial checker could then connect their abstract statements to the emitted coefficient files. The direct orientation lemma needs a formal signed-Pell classification and the exact source frame bound. Closing these interfaces would strengthen the verification status without changing the literal equations.

Comparisons under a stated objective. It remains reasonable to minimize arity subject to degree four and height one, or minimize support subject to a fixed arity cap. These are concrete optimization problems. A single unqualified phrase such as “the simplest Diophantine equation” does not select between such incomparable outcomes. The present release makes progress under explicit criteria and leaves the global question open.

A Literal direct residuals

Every coordinate in this appendix ranges independently over $\mathbb{Z}$. The single polynomial is the sum of the squares of *all* the displayed residuals, not the square of their sum. Coordinate names impose no extra domain restrictions. This transcription is generated from the same literal JSON checked by the independent expansion verifier.

0: multiplication

$$r_0 = -z_3^2 + z_4$$

1: multiplication

$$r_1 = -1 - z_0 z_2 + z_9$$

2: multiplication

$$r_2 = -1 + z_1 + z_2 - z_2 z_7 - z_3 - z_7$$

3: multiplication

$$r_3 = -2 + z_1 - z_8 z_9$$

4: multiplication

$$r_4 = 1 + z_4 - z_5 z_{20} + z_{18}$$

5: multiplication

$$r_5 = -1 + 2z_{10} - z_{10}^2 - 2z_{10}z_{11} + 2z_{11} - z_{11}^2 + z_{21}$$

6: multiplication

$$\begin{aligned} r_6 = {} & -5 - 30z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 24z_{10}z_{21} - 40z_{11} - 64z_{11}^2 - 32z_{11}z_{21} - 10z_{21} - 4z_{21}^2 \\ & + z_{22} \end{aligned}$$

7: multiplication

$$r_7 = -2 + 2z_{10} - z_{10}^2 - 4z_{10}z_{22} + 4z_{22} - 4z_{22}^2 + z_{23}$$

8: multiplication

$$r_8 = 1 + 6z_{10} - 36z_{10}^2 - 192z_{10}z_{22} - 24z_{10}z_{23} + 16z_{22} - 256z_{22}^2 - 64z_{22}z_{23} + 2z_{23} - 4z_{23}^2 + z_{24}$$

9: multiplication

$$r_9 = -2 - z_{11}^2 - 2z_{11}z_{12} - z_{12}^2 + z_{25}$$

10: multiplication

$$r_{10} = -9 + 7z_{10} - z_{10}^2 - 4z_{10}z_{11} - 4z_{10}z_{25} + 12z_{11} - 4z_{11}^2 - 8z_{11}z_{25} + 12z_{25} - 4z_{25}^2 + z_{26}$$

11: multiplication

$$r_{11} = -1 + 2z_{10} - z_{10}^2 - 2z_{10}z_{12} + 2z_{12} - z_{12}^2 + z_{27}$$

12: multiplication

$$\begin{aligned} r_{12} = {} & -6z_{10} - 144z_{10}^2 - 192z_{10}z_{11} - 192z_{10}z_{12} - 48z_{10}z_{21} - 48z_{10}z_{27} - 8z_{11} - 64z_{11}^2 \\ & - 128z_{11}z_{12} - 32z_{11}z_{21} - 32z_{11}z_{27} - 64z_{12}^2 - 32z_{12}z_{21} - 32z_{12}z_{27} - 2z_{21} - 4z_{21}^2 \\ & - 8z_{21}z_{27} - 4z_{27}^2 + z_{28} \end{aligned}$$

13: multiplication

$$r_{13} = -4z_{26}^2 - 8z_{26}z_{28} - 4z_{28}^2 + z_{29}$$

14: multiplication

$$r_{14} = -6 + 16z_{12} - 16z_{12}^2 - 16z_{12}z_{25} - 16z_{12}z_{27} + 8z_{25} - 4z_{25}^2 - 8z_{25}z_{27} + 8z_{27} - 4z_{27}^2 + z_{30}$$

15: multiplication

$$\begin{aligned} r_{15} = {} & 1 + 6z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 48z_{10}z_{12} - 24z_{10}z_{21} - 48z_{10}z_{27} - 24z_{10}z_{30} + 8z_{11} - 64z_{11}^2 \\ & - 64z_{11}z_{12} - 32z_{11}z_{21} - 64z_{11}z_{27} - 32z_{11}z_{30} + 8z_{12} - 16z_{12}^2 - 16z_{12}z_{21} - 32z_{12}z_{27} \\ & - 16z_{12}z_{30} + 2z_{21} - 4z_{21}^2 - 16z_{21}z_{27} - 8z_{21}z_{30} + 8z_{27} - 16z_{27}^2 - 16z_{27}z_{30} + 4z_{30} \\ & - 4z_{30}^2 + z_{31} \end{aligned}$$

16: multiplication

$$r_{16} = -10 - 5z_{10} - z_{10}^2 - 12z_{10}z_{11} - 16z_{10}z_{12} - 4z_{10}z_{25} - 36z_{11} - 36z_{11}^2 - 96z_{11}z_{12} - 24z_{11}z_{25} - 48z_{12} - 64z_{12}^2 - 32z_{12}z_{25} - 12z_{25} - 4z_{25}^2 + z_{32}$$

17: multiplication

$$r_{17} = -2 - 24z_{10} - 36z_{10}^2 - 12z_{10}z_{11} - 96z_{10}z_{12} - 24z_{10}z_{27} - 3z_{11} - z_{11}^2 - 16z_{11}z_{12} - 4z_{11}z_{27} - 32z_{12} - 64z_{12}^2 - 32z_{12}z_{27} - 8z_{27} - 4z_{27}^2 + z_{33}$$

18: multiplication

$$r_{18} = 2 - 4z_{32}^2 - 8z_{32}z_{33} - 4z_{33}^2 + z_{34}$$

19: multiplication

$$r_{19} = -12 + 16z_{10} - 4z_{10}^2 - 8z_{10}z_{12} - 8z_{10}z_{21} - 8z_{10}z_{25} + 16z_{12} - 4z_{12}^2 - 8z_{12}z_{21} - 8z_{12}z_{25} + 16z_{21} - 4z_{21}^2 - 8z_{21}z_{25} + 16z_{25} - 4z_{25}^2 + z_{35}$$

20: multiplication

$$r_{20} = -3 - 24z_{10} - 64z_{10}^2 - 160z_{10}z_{12} - 64z_{10}z_{25} - 32z_{10}z_{27} - 32z_{10}z_{35} - 26z_{12} - 100z_{12}^2 - 80z_{12}z_{25} - 40z_{12}z_{27} - 40z_{12}z_{35} - 8z_{25} - 16z_{25}^2 - 16z_{25}z_{27} - 16z_{25}z_{35} - 6z_{27} - 4z_{27}^2 - 8z_{27}z_{35} - 4z_{35} - 4z_{35}^2 + z_{36}$$

21: multiplication

$$r_{21} = -1 - 18z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 24z_{10}z_{21} - 24z_{11} - 64z_{11}^2 - 32z_{11}z_{21} - 6z_{21} - 4z_{21}^2 + z_{37}$$

22: multiplication

$$r_{22} = -2 + 2z_{10} - z_{10}^2 - 4z_{10}z_{37} + 4z_{37} - 4z_{37}^2 + z_{38}$$

23: multiplication

$$r_{23} = 1 + 6z_{10} - 36z_{10}^2 - 192z_{10}z_{37} - 24z_{10}z_{38} + 16z_{37} - 256z_{37}^2 - 64z_{37}z_{38} + 2z_{38} - 4z_{38}^2 + z_{39}$$

24: multiplication

$$r_{24} = -z_{11}^2 - 4z_{11}z_{39} - 4z_{39}^2 + z_{40}$$

25: multiplication

$$r_{25} = -6 - 30z_{10} - 64z_{10}^2 - 128z_{10}z_{11} - 64z_{10}z_{21} - 32z_{11} - 64z_{11}^2 - 64z_{11}z_{21} - 14z_{21} - 16z_{21}^2 + z_{41}$$

26: multiplication

$$r_{26} = -4 + 8z_{11} - 64z_{11}^2 - 96z_{11}z_{12} - 32z_{11}z_{25} - 32z_{11}z_{41} + 6z_{12} - 36z_{12}^2 - 24z_{12}z_{25} - 24z_{12}z_{41} + 2z_{25} - 4z_{25}^2 - 8z_{25}z_{41} + 4z_{41} - 4z_{41}^2 + z_{42}$$

27: multiplication

$$r_{27} = 8z_{10} - 64z_{10}^2 - 96z_{10}z_{12} - 32z_{10}z_{27} - 32z_{10}z_{42} + 6z_{12} - 36z_{12}^2 - 24z_{12}z_{27} - 24z_{12}z_{42} + 2z_{27} - 4z_{27}^2 - 8z_{27}z_{42} + 4z_{42} - 4z_{42}^2 + z_{43}$$

28: multiplication

$$r_{28} = -z_{12}^2 - 2z_{12}z_{13} - z_{13}^2 + z_{44}$$

29: multiplication

$$r_{29} = 2z_{10} - z_{10}^2 - 4z_{10}z_{12} - 4z_{10}z_{44} + 4z_{12} - 4z_{12}^2 - 8z_{12}z_{44} + 4z_{44} - 4z_{44}^2 + z_{45}$$

30: multiplication

$$r_{30} = -35 + 24z_{10} - 4z_{10}^2 - 8z_{10}z_{11} - 8z_{10}z_{25} - 8z_{10}z_{27} + 24z_{11} - 4z_{11}^2 - 8z_{11}z_{25} - 8z_{11}z_{27} + 24z_{25} - 4z_{25}^2 - 8z_{25}z_{27} + 24z_{27} - 4z_{27}^2 + z_{46}$$

31: multiplication

$$r_{31} = -16z_{10}^2 - 32z_{10}z_{12} - 32z_{10}z_{27} - 32z_{10}z_{44} - 16z_{10}z_{45} - 16z_{10}z_{46} - 4z_{12} - 16z_{12}^2 - 32z_{12}z_{27} - 32z_{12}z_{44} - 16z_{12}z_{45} - 16z_{12}z_{46} - 16z_{27}^2 - 32z_{27}z_{44} - 16z_{27}z_{45} - 16z_{27}z_{46} - 4z_{44} - 16z_{44}^2 - 16z_{44}z_{45} - 16z_{44}z_{46} - 2z_{45} - 4z_{45}^2 - 8z_{45}z_{46} - 4z_{46}^2 + z_{47}$$

32: multiplication

$$r_{32} = z_{12} - 25z_{12}^2 - 40z_{12}z_{27} - 20z_{12}z_{30} - 16z_{27}^2 - 16z_{27}z_{30} - 4z_{30}^2 + z_{48}$$

33: multiplication

$$r_{33} = -1 + 8z_{11} - 4z_{11}^2 - 8z_{11}z_{25} + 8z_{25} - 4z_{25}^2 + z_{49}$$

34: multiplication

$$r_{34} = -1 - 16z_{11}^2 - 8z_{11}z_{12} - 32z_{11}z_{25} - 16z_{11}z_{49} - z_{12}^2 - 8z_{12}z_{25} - 4z_{12}z_{49} - 16z_{25}^2 - 16z_{25}z_{49} - 4z_{49}^2 + z_{50}$$

35: multiplication

$$r_{35} = 1 + z_{11} - z_{11}^2 - 4z_{11}z_{12} - 4z_{11}z_{50} - 4z_{12}^2 - 8z_{12}z_{50} - 4z_{50}^2 + z_{51}$$

36: multiplication

$$r_{36} = -1 - 4z_{51} - 4z_{51}^2 + z_{52}$$

37: multiplication

$$r_{37} = -1 - z_{12}^2 - 8z_{12}z_{51} - 4z_{12}z_{52} - 16z_{51}^2 - 16z_{51}z_{52} - 4z_{52}^2 + z_{53}$$

38: multiplication

$$r_{38} = 2 + z_{11} - z_{11}^2 - 4z_{11}z_{12} - 4z_{11}z_{53} - 4z_{12}^2 - 8z_{12}z_{53} - 4z_{53}^2 + z_{54}$$

39: multiplication

$$r_{39} = -4z_{48}^2 - 8z_{48}z_{54} - 4z_{54}^2 + z_{55}$$

40: multiplication

$$r_{40} = -9 + 7z_{10} - z_{10}^2 - 8z_{10}z_{48} - 4z_{10}z_{55} + 24z_{48} - 16z_{48}^2 - 16z_{48}z_{55} + 12z_{55} - 4z_{55}^2 + z_{56}$$

41: multiplication

$$r_{41} = -1 - 4z_{56} - 4z_{56}^2 + z_{57}$$

42: multiplication

$$r_{42} = -z_{12}^2 - 8z_{12}z_{56} - 4z_{12}z_{57} - 16z_{56}^2 - 16z_{56}z_{57} - 4z_{57}^2 + z_{58}$$

43: multiplication

$$r_{43} = -z_{10} - z_{10}^2 - 4z_{10}z_{12} - 4z_{10}z_{58} - 4z_{12} - 4z_{12}^2 - 8z_{12}z_{58} - 4z_{58} - 4z_{58}^2 + z_{59}$$

44: multiplication

$$r_{44} = -10 - 12z_{59} - 4z_{59}^2 + z_{60}$$

45: multiplication

$$r_{45} = 1 + 4z_{11} - 4z_{11}^2 - 8z_{11}z_{25} - 16z_{11}z_{59} - 8z_{11}z_{60} + 4z_{25} - 4z_{25}^2 - 16z_{25}z_{59} - 8z_{25}z_{60} + 8z_{59} - 16z_{59}^2 - 16z_{59}z_{60} + 4z_{60} - 4z_{60}^2 + z_{61}$$

46: multiplication

$$r_{46} = z_{11} - 25z_{11}^2 - 40z_{11}z_{25} - 20z_{11}z_{61} - 16z_{25}^2 - 16z_{25}z_{61} - 4z_{61}^2 + z_{62}$$

47: multiplication

$$r_{47} = -10 - 12z_{54} - 4z_{54}^2 + z_{63}$$

48: multiplication

$$r_{48} = 1 - 16z_{54}^2 - 16z_{54}z_{62} - 16z_{54}z_{63} - 4z_{62}^2 - 8z_{62}z_{63} - 4z_{63}^2 + z_{64}$$

49: multiplication

$$r_{49} = z_{12} - z_{12}^2 - 8z_{12}z_{62} - 4z_{12}z_{64} - 16z_{62}^2 - 16z_{62}z_{64} - 4z_{64}^2 + z_{65}$$

50: multiplication

$$r_{50} = -4z_{10} - 4z_{10}^2 - 4z_{10}z_{12} - 8z_{10}z_{21} - z_{12} - z_{12}^2 - 4z_{12}z_{21} - 4z_{21} - 4z_{21}^2 + z_{66}$$

51: multiplication

$$r_{51} = 1 + 4z_{10} - 4z_{10}^2 - 8z_{10}z_{21} - 8z_{10}z_{66} + 4z_{21} - 4z_{21}^2 - 8z_{21}z_{66} + 4z_{66} - 4z_{66}^2 + z_{67}$$

52: multiplication

$$r_{52} = -2 - 4z_{10} - 16z_{10}^2 - 32z_{10}z_{21} - 16z_{10}z_{65} - 16z_{10}z_{67} - 4z_{21} - 16z_{21}^2 - 16z_{21}z_{65} - 16z_{21}z_{67} - 4z_{65}^2 - 8z_{65}z_{67} - 2z_{67} - 4z_{67}^2 + z_{68}$$

53: multiplication

$$r_{53} = -2 - 4z_{68} - 4z_{68}^2 + z_{69}$$

54: multiplication

$$r_{54} = -z_3 z_{15} + z_{70}$$

55: multiplication

$$r_{55} = z_{14} - z_{15} - 2z_{24}z_{70} - z_{71}$$

56: multiplication

$$r_{56} = z_{14} - z_{15} - 4z_{26}z_{70} - 2z_{29}z_{70} - z_{72}$$

57: multiplication

$$r_{57} = z_{14} - z_{15} - 2z_{31}z_{70} - 6z_{70} - z_{73}$$

58: multiplication

$$r_{58} = z_{14} - z_{15} - 4z_{32}z_{70} - 2z_{34}z_{70} - z_{74}$$

59: multiplication

$$r_{59} = z_{14} - z_{15} - 2z_{36}z_{70} + 6z_{70} - z_{75}$$

60: multiplication

$$r_{60} = z_{14} - z_{15} - 4z_{39}z_{70} - 2z_{40}z_{70} - z_{76}$$

61: multiplication

$$r_{61} = z_{14} - z_{15} - 2z_{43}z_{70} - z_{77}$$

62: multiplication

$$r_{62} = z_{14} - z_{15} - 2z_{47}z_{70} - 4z_{70} - z_{78}$$

63: nonnegative_square_difference

$$r_{63} = -z_{10}z_{11} + z_{10}z_{12} + z_{11} + z_{11}z_{12} - z_{12} - z_{12}^2 + z_{79}$$

64: multiplication

$$r_{64} = 1 + 2z_{13} + z_{13}^2 - z_{79}^2 + z_{80}$$

65: nonnegative_square_difference

$$r_{65} = -z_{80}^2 + z_{81}$$

66: multiplication

$$r_{66} = -4z_{68}z_{70} - 2z_{69}z_{70} + 2z_{70} + z_{82}$$

67: nonnegative_square_difference

$$r_{67} = -z_{14}^2 + 2z_{14}z_{15} + 2z_{14}z_{82} - z_{15}^2 - 2z_{15}z_{82} - z_{82}^2 + z_{83}$$

68: multiplication

$$r_{68} = -1 - z_3 z_{11} + z_{84}$$

69: multiplication

$$r_{69} = -z_{15}z_{84} + z_{86}z_{87} + z_{88}$$

70: multiplication

$$r_{70} = -1 + z_3 - z_3 z_{10} + z_{85}$$

71: multiplication

$$r_{71} = -1 + 2z_3 - 2z_3 z_{10} - 2z_3 z_{21} + z_{86}$$

72: multiplication

$$r_{72} = -z_{14}z_{85} + z_{87}$$

73: multiplication

$$r_{73} = -1 - 6z_3 z_{10} - 8z_3 z_{11} - 2z_3 z_{21} + z_{89}$$

74: multiplication

$$r_{74} = z_{14}z_{84} - z_{15}z_{89} + z_{90}$$

75: multiplication

$$r_{75} = -z_{14}z_{71} + z_{15}z_{71} + z_{91}$$

76: multiplication

$$r_{76} = -z_{72}z_{91} + z_{92}$$

77: multiplication

$$r_{77} = -z_{73}z_{92} + z_{93}$$

78: multiplication

$$r_{78} = -z_{74}z_{93} + z_{94}$$

79: multiplication

$$r_{79} = -z_{75}z_{94} + z_{95}$$

80: multiplication

$$r_{80} = -z_{76}z_{95} + z_{96}$$

81: multiplication

$$r_{81} = -z_{77}z_{96} + z_{97}$$

82: multiplication

$$r_{82} = -z_{78}z_{97} + z_{98}$$

83: multiplication

$$r_{83} = -z_{88}z_{98} + z_{99}$$

84: multiplication

$$r_{84} = -z_{90}z_{99} + z_{100}$$

85: multiplication

$$r_{85} = -z_{100}^2 + z_{101}$$

86: multiplication

$$r_{86} = -1 - z_2 - z_2z_{18} + z_{102}$$

87: multiplication

$$r_{87} = -1 - z_{16}z_{102} + z_{103}$$

88: nonnegative_square_difference

$$\begin{aligned} r_{88} = &-2z_1^2 + 2z_1z_{14} + 2z_1z_{15} + 2z_1z_{103} + 2z_1z_{104} - z_{14}^2 - 2z_{14}z_{103} - z_{15}^2 - 2z_{15}z_{104} \\ &+ z_{19}z_{106} - z_{81}z_{101} - z_{83}z_{101} - z_{103}^2 - z_{104}^2 \end{aligned}$$

89: multiplication

$$r_{89} = -1 - z_2z_{17} - z_{17}z_{102} + z_{104}$$

90: addition

$$\begin{aligned} r_{90} = &-9z_{10} - 9z_{11} - 4z_{21} - 2z_{24} - 4z_{26} - 2z_{29} - 2z_{31} - 4z_{32} - 2z_{34} - 2z_{36} - 4z_{39} - 2z_{40} - 2z_{43} - 2z_{47} \\ &- 4z_{68} - 2z_{69} + z_{105} \end{aligned}$$

91: multiplication

$$r_{91} = -z_{107}z_{110} + z_{110} + z_{112}$$

92: multiplication

$$r_{92} = -z_5z_{109} + z_{108} - z_{109}z_{112} - z_{111}$$

93: multiplication

$$r_{93} = -z_{19}z_{114} + z_{117}$$

94: multiplication

$$r_{94} = z_{113} - z_{115}z_{117} - z_{116}$$

95: multiplication

$$r_{95} = -z_{117}z_{119} + z_{118} - z_{120}$$

96: multiplication

$$r_{96} = -z_{117}z_{122} + z_{121} - z_{123}$$

97: multiplication

$$r_{97} = -z_{117}z_{125} + z_{124} - z_{126}$$

98: multiplication

$$r_{98} = -z_{19}z_{128} + z_{131}$$

99: multiplication

$$r_{99} = z_{127} - z_{129}z_{131} - z_{130}$$

100: multiplication

$$r_{100} = -z_{131}z_{133} + z_{132} - z_{134}$$

101: multiplication

$$r_{101} = -z_{19}z_{136} + z_{139}$$

102: multiplication

$$r_{102} = z_{135} - z_{137}z_{139} - z_{138}$$

103: multiplication

$$r_{103} = -z_{139}z_{141} + z_{140} - z_{142}$$

104: multiplication

$$r_{104} = -z_{19}z_{109} + z_{146}$$

105: multiplication

$$r_{105} = z_{143} - z_{144}z_{146} - z_{145}$$

106: multiplication

$$r_{106} = -z_{117}z_{148} + z_{147} - z_{149}$$

107: multiplication

$$r_{107} = z_{150} - 2z_{150}z_{152} + z_{153}$$

108: multiplication

$$r_{108} = -z_{150}z_{153} - z_{153} + z_{154}$$

109: shared.last_norm

$$r_{109} = 2z_{151} + z_{151}^2 + z_{153}^2 - z_{154}^2$$

110: multiplication

$$r_{110} = 1 + z_{151} - z_{151}z_{156} - z_{156} + z_{157}$$

111: index0.norm1

$$r_{111} = z_{155}^2 - 2z_{156} + z_{156}^2 - z_{157}^2$$

112: multiplication

$$r_{112} = z_4^2 - 2z_4z_{151} - 2z_{151} + z_{158}$$

113: multiplication

$$r_{113} = 1 + z_4 - z_4z_{156} + z_6 - z_{155} - z_{156} + z_{157} + z_{158}z_{159}$$

114: multiplication

$$r_{114} = -z_{151}z_{161} - z_{161} + z_{162}$$

115: index1.norm1

$$r_{115} = -1 + z_{160}^2 + z_{161}^2 - z_{162}^2$$

116: multiplication

$$r_{116} = 1 - 2z_5 + z_5^2 - 2z_5z_{151} + z_{163}$$

117: multiplication

$$r_{117} = -z_5z_{161} + z_{19} - z_{160} + z_{162} + z_{163}z_{164}$$

118: base2.T_definition

$$r_{118} = -1 + 2z_{151}z_{224} - z_{165} + 2z_{224} - z_{224}^2$$

119: multiplication

$$r_{119} = -1 + z_{107} - z_{160} - z_{161}z_{224} + z_{162} + z_{165}z_{166}$$

120: multiplication

$$r_{120} = 1 + z_{151} - z_{151}z_{168} - z_{168} + z_{169}$$

121: index2.norm1

$$r_{121} = z_{167}^2 - 2z_{168} + z_{168}^2 - z_{169}^2$$

122: multiplication

$$r_{122} = 1 - 2z_{107} + z_{107}^2 - 2z_{107}z_{151} + z_{170}$$

123: multiplication

$$r_{123} = z_{107} - z_{107}z_{168} + z_{108} - z_{167} + z_{169} + z_{170}z_{171}$$

124: multiplication

$$r_{124} = 4 - 4z_{107} + z_{107}^2 - 2z_{107}z_{151} + 2z_{151} + z_{172}$$

125: multiplication

$$r_{125} = -1 + z_{107} - z_{107}z_{156} + z_{109} - z_{155} + z_{156} + z_{157} + z_{172}z_{173}$$

126: multiplication

$$r_{126} = 1 + z_{151} - z_{151}z_{175} - z_{175} + z_{176}$$

127: index3.norm1

$$r_{127} = z_{174}^2 - 2z_{175} + z_{175}^2 - z_{176}^2$$

128: multiplication

$$r_{128} = z_{107} - z_{107}z_{175} + z_{113} + z_{170}z_{177} - z_{174} + z_{176}$$

129: multiplication

$$r_{129} = -2z_{151} - z_{151}^2 + z_{178}$$

130: multiplication

$$r_{130} = -z_{107}z_{155} - z_{107}z_{157} + z_{114} + z_{151}z_{157} + z_{155} - z_{156}z_{178} + 2z_{157} + z_{172}z_{179} + z_{178}$$

131: multiplication

$$r_{131} = -z_{151}z_{181} - z_{181} + z_{182}$$

132: index5.norm1

$$r_{132} = -1 + z_{180}^2 + z_{181}^2 - z_{182}^2$$

133: multiplication

$$r_{133} = -z_{107}z_{181} + z_{118} + z_{170}z_{183} - z_{180} + z_{182}$$

134: multiplication

$$r_{134} = -z_{151}z_{185} - z_{185} + z_{186}$$

135: index6.norm1

$$r_{135} = -1 + z_{184}^2 + z_{185}^2 - z_{186}^2$$

136: multiplication

$$r_{136} = -z_{107}z_{185} + z_{121} + z_{170}z_{187} - z_{184} + z_{186}$$

137: multiplication

$$r_{137} = -z_{151}z_{189} - z_{189} + z_{190}$$

138: index7.norm1

$$r_{138} = -1 + z_{188}^2 + z_{189}^2 - z_{190}^2$$

139: multiplication

$$r_{139} = -z_{107}z_{189} + z_{124} + z_{170}z_{191} - z_{188} + z_{190}$$

140: multiplication

$$r_{140} = -z_{151}z_{193} - z_{193} + z_{194}$$

141: index8.norm1

$$r_{141} = -1 + z_{192}^2 + z_{193}^2 - z_{194}^2$$

142: multiplication

$$r_{142} = -z_{107}z_{193} + z_{127} + z_{170}z_{195} - z_{192} + z_{194}$$

143: multiplication

$$r_{143} = -1 - z_{151} - z_{151}z_{197} - z_{197} + z_{198}$$

144: index9.norm1

$$r_{144} = z_{196}^2 + 2z_{197} + z_{197}^2 - z_{198}^2$$

145: multiplication

$$r_{145} = 1 - z_{107} - z_{107}z_{197} + z_{128} + z_{172}z_{199} - z_{196} + z_{197} + z_{198}$$

146: multiplication

$$r_{146} = -z_{151}z_{201} - z_{201} + z_{202}$$

147: index10.norm1

$$r_{147} = -1 + z_{200}^2 + z_{201}^2 - z_{202}^2$$

148: multiplication

$$r_{148} = -z_{107}z_{201} + z_{132} + z_{170}z_{203} - z_{200} + z_{202}$$

149: multiplication

$$r_{149} = -z_{151}z_{205} - z_{205} + z_{206}$$

150: index11.norm1

$$r_{150} = -1 + z_{204}^2 + z_{205}^2 - z_{206}^2$$

151: multiplication

$$r_{151} = -z_{107}z_{205} + z_{135} + z_{170}z_{207} - z_{204} + z_{206}$$

152: multiplication

$$r_{152} = -z_{151}z_{209} - z_{209} + z_{210}$$

153: index12.norm1

$$r_{153} = -1 + z_{208}^2 + z_{209}^2 - z_{210}^2$$

154: multiplication

$$r_{154} = -z_{107}z_{209} + z_{136} + z_{172}z_{211} - z_{208} + z_{209} + z_{210}$$

155: multiplication

$$r_{155} = -z_{151}z_{213} - z_{213} + z_{214}$$

156: index13.norm1

$$r_{156} = -1 + z_{212}^2 + z_{213}^2 - z_{214}^2$$

157: multiplication

$$r_{157} = -z_{107}z_{213} + z_{140} + z_{170}z_{215} - z_{212} + z_{214}$$

158: multiplication

$$r_{158} = -z_{151}z_{217} - z_{217} + z_{218}$$

159: index14.norm1

$$r_{159} = -1 + z_{216}^2 + z_{217}^2 - z_{218}^2$$

160: multiplication

$$r_{160} = -z_{107}z_{217} + z_{143} + z_{170}z_{219} - z_{216} + z_{218}$$

161: multiplication

$$r_{161} = -z_{151}z_{221} - z_{221} + z_{222}$$

162: index15.norm1

$$r_{162} = -1 + z_{220}^2 + z_{221}^2 - z_{222}^2$$

163: multiplication

$$r_{163} = -z_{107}z_{221} + z_{147} + z_{170}z_{223} - z_{220} + z_{222}$$

164: shared.frame_definition

$$\begin{aligned} r_{164} = & -8 + 4z_0 - 7z_0^2 - 8z_0z_1 - 8z_0z_2 - 8z_0z_3 - 2z_0z_4 - 5z_1^2 - 8z_1z_2 - 8z_1z_3 - 2z_2 \\ & - 5z_2^2 - 8z_2z_3 - 4z_3^2 - 2z_4^2 - z_5^2 - z_6^2 + 2z_{10} - z_{10}^2 - z_{11}^2 - z_{12}^2 - z_{13}^2 \\ & - z_{14}^2 - z_{15}^2 - z_{16}^2 - z_{17}^2 - z_{18}^2 - z_{19}^2 - z_{105}^2 + 2z_{107} - 2z_{107}^2 - z_{108}^2 \\ & - z_{109}^2 - z_{113}^2 - z_{114}^2 - z_{118}^2 - z_{121}^2 - z_{124}^2 - z_{127}^2 - z_{128}^2 - z_{132}^2 \\ & - z_{135}^2 - z_{136}^2 - z_{140}^2 - z_{143}^2 - z_{147}^2 + z_{150} - z_{224}^2 \end{aligned}$$

165: self_radix.base_definition

$$r_{165} = -z_5z_{10} - z_5z_{11} - z_5z_{12} - z_5z_{13} - z_5z_{14} - z_5z_{15} - z_5z_{16} - z_5z_{17} - z_5z_{18} + z_{224}$$

166: cofinal.horizon

$$r_{166} = -17 + z_0 - z_{225}^2$$

167: cofinal.beta_base

$$r_{167} = z_2 + z_6 - z_6^2$$

168: cap.main_Ay

$$r_{168} = -z_{151}z_{227} - z_{227} + z_{238}$$

169: cap.main_norm

$$r_{169} = -1 + z_{226}^2 + z_{227}^2 - z_{238}^2$$

170: cap.common_Ay

$$r_{170} = -z_{151}z_{229} - z_{229} + z_{239}$$

171: cap.common_norm

$$r_{171} = -1 + z_{228}^2 + z_{229}^2 - z_{239}^2$$

172: cap.BR

$$r_{172} = -z_{231} - z_{231}z_{232} + z_{240}$$

173: cap.comparison_norm

$$r_{173} = -1 + z_{230}^2 + z_{231}^2 - z_{240}^2$$

174: cap.Y2

$$r_{174} = -z_{227}^2 + z_{241}$$

175: cap.common_multiple

$$r_{175} = z_{229} - z_{234}z_{241} - z_{241}$$

176: cap.comparison_constructor

$$r_{176} = -4z_{229} - 8z_{229}z_{233} + z_{232}$$

177: cap.comparison_modulus

$$r_{177} = -z_{151} - z_{228}z_{235} + z_{232}$$

178: cap.abscissa_residue

$$r_{178} = -z_{226} - z_{228}z_{236} + z_{230}$$

179: cap.index_residue

$$r_{179} = -z_{150} - 4z_{227}z_{237} + z_{231}$$

180: cap.pooled_index_residue

$$\begin{aligned} r_{180} = & 9 + 6z_0^2 + 8z_0z_1 + 8z_0z_2 + 8z_0z_3 + 2z_0z_4 - 2z_0z_{156} - 4z_0z_{161} - 2z_0z_{168} + 5z_1^2 \\ & + 8z_1z_2 + 8z_1z_3 - 4z_1z_{161} - 2z_1z_{209} + 5z_2^2 + 8z_2z_3 - 4z_2z_{161} - 2z_2z_{197} + 4z_3^2 \\ & - 4z_3z_{161} + z_4^2 - 2z_4z_{168} + z_{10}^2 - 2z_{10}z_{175} + z_{11}^2 - 2z_{11}z_{181} + z_{12}^2 - 2z_{12}z_{185} \\ & + z_{13}^2 - 2z_{13}z_{189} + z_{14}^2 - 2z_{14}z_{193} + z_{15}^2 - 2z_{15}z_{201} + z_{16}^2 - 2z_{16}z_{205} + z_{17}^2 \\ & - 2z_{17}z_{213} + z_{18}^2 - 2z_{18}z_{217} + z_{105}^2 - 2z_{105}z_{221} - z_{151}z_{242} + z_{156}^2 + z_{161}^2 \\ & + z_{168}^2 - z_{174} + z_{175}^2 - z_{180} + z_{181}^2 - z_{184} + z_{185}^2 - z_{188} + z_{189}^2 - z_{192} + z_{193}^2 \\ & + z_{197}^2 - z_{200} + z_{201}^2 - z_{204} + z_{205}^2 + z_{209}^2 - z_{212} + z_{213}^2 - z_{216} + z_{217}^2 + z_{221}^2 \end{aligned}$$

181: domain_guard.0

$$r_{181} = -1 + z_1 - z_{243}^2 - z_{244}^2 - z_{245} - z_{245}^2$$

182: domain_guard.1

$$r_{182} = -1 - z_3 z_5 + z_5 + z_6 - z_{246}^2 - z_{247}^2 - z_{248} - z_{248}^2$$

183: domain_guard.2

$$r_{183} = z_3 z_5 - z_6 - z_{249}^2 - z_{250}^2 - z_{251} - z_{251}^2$$

184: domain_guard.12

$$r_{184} = -2 - z_5 + z_{107} - z_{252}^2 - z_{253}^2 - z_{254} - z_{254}^2$$

185: domain_guard.13

$$r_{185} = 1 + z_{151} - z_{255}^2 - z_{256}^2 - z_{257} - z_{257}^2$$

186: domain_guard.14

$$r_{186} = z_{109} z_{111} - z_{111} - z_{111}^2 - z_{258}^2 - z_{259}^2 - z_{260} - z_{260}^2$$

187: domain_guard.15

$$r_{187} = z_{114} z_{116} - z_{116} - z_{116}^2 - z_{261}^2 - z_{262}^2 - z_{263} - z_{263}^2$$

188: domain_guard.16

$$r_{188} = z_{114} z_{120} - z_{120} - z_{120}^2 - z_{264}^2 - z_{265}^2 - z_{266} - z_{266}^2$$

189: domain_guard.17

$$r_{189} = z_{114} z_{123} - z_{123} - z_{123}^2 - z_{267}^2 - z_{268}^2 - z_{269} - z_{269}^2$$

190: domain_guard.18

$$r_{190} = z_{114} z_{126} - z_{126} - z_{126}^2 - z_{270}^2 - z_{271}^2 - z_{272} - z_{272}^2$$

191: domain_guard.19

$$r_{191} = z_{128} z_{130} - z_{130} - z_{130}^2 - z_{273}^2 - z_{274}^2 - z_{275} - z_{275}^2$$

192: domain_guard.20

$$r_{192} = z_{128} z_{134} - z_{134} - z_{134}^2 - z_{276}^2 - z_{277}^2 - z_{278} - z_{278}^2$$

193: domain_guard.21

$$r_{193} = z_{136} z_{138} - z_{138} - z_{138}^2 - z_{279}^2 - z_{280}^2 - z_{281} - z_{281}^2$$

194: domain_guard.22

$$r_{194} = z_{136} z_{142} - z_{142} - z_{142}^2 - z_{282}^2 - z_{283}^2 - z_{284} - z_{284}^2$$

195: domain_guard.23

$$r_{195} = z_{109} z_{145} - z_{145} - z_{145}^2 - z_{285}^2 - z_{286}^2 - z_{287} - z_{287}^2$$

196: domain_guard.24

$$r_{196} = z_{114} z_{149} - z_{149} - z_{149}^2 - z_{288}^2 - z_{289}^2 - z_{290} - z_{290}^2$$

197: domain_guard.25

$$r_{197} = z_{234} - z_{291}^2 - z_{292}^2 - z_{293} - z_{293}^2$$

198: domain_guard.26

$$\begin{aligned} r_{198} = & -5 - z_{150} + 2z_{156} - z_{156}^2 - z_{161}^2 + 2z_{168} - z_{168}^2 + 2z_{175} - z_{175}^2 - z_{181}^2 - z_{185}^2 \\ & - z_{189}^2 - z_{193}^2 - 2z_{197} - z_{197}^2 - z_{201}^2 - z_{205}^2 - z_{209}^2 - z_{213}^2 - z_{217}^2 - z_{221}^2 \\ & + z_{227} - z_{294}^2 - z_{295}^2 - z_{296} - z_{296}^2 \end{aligned}$$

199: orientation.product_2

$$r_{199} = -z_{10} z_{11} + z_{11} + z_{297}$$

200: orientation.product_3

$$r_{200} = -z_{12} z_{297} + z_{298}$$

201: orientation.product_4

$$r_{201} = -z_{13} z_{298} + z_{299}$$

202: orientation.product_5

$$r_{202} = -z_{14}z_{299} + z_{300}$$

203: orientation.product_6

$$r_{203} = -z_{15}z_{300} + z_{301}$$

204: orientation.product_7

$$r_{204} = -z_{16}z_{301} + z_{302}$$

205: orientation.fused_strict_product

$$r_{205} = -1 + z_{17}z_{302} - z_{303}^2 - z_{304}^2 - z_{305} - z_{305}^2$$

B Source identities and comparison inventory

The comparison and retrieval pin is `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. The direct parent comes from `131-shared-pell-caps-round16/direct/integer324/`; the matched graph parent comes from `127-graph-certificates/graph/principal/`. The 306-coordinate source is newly constructed here. The complete per-file inventory is `MANIFEST.sha256`; this appendix records the normative equation identities.

`direct/parent/integer324-equation.json` 36,840 bytes. SHA-256: `32618060c22765f54423b53f9faef437d736494da82aef309efe96210d6cbadd`.

`direct/integer306/equation.json` 35,585 bytes. SHA-256: `d51d75d2c6398fa3410b19ae43ba8d80f86c668ad670a803f6777a2af221c189`.

`direct/integer306/expanded_polynomial.json` 114,236 bytes. SHA-256: `c3b4b6c7646ba6a92e02aa24fc9d64e62d642e47046a6d38c1e01df7357d293a`.

`graph/matched1252/polynomial.json` 656,104 bytes. SHA-256: `66a29766c5b1107c36674c38215b02afaba1802606cb4f1a0584cc24302c2bf8`.

`graph/principal985/polynomial.json` 1,984,728 bytes. SHA-256: `33e35bb2f0354cbcc47bac27056c8fb560c508cb78ce04b1bb1dff9adcf7a635`.

`graph/sparse1122/polynomial.json` 580,440 bytes. SHA-256: `69edd5df14b21dee3960da9cb86f5723e0e99aaf13814922cda965493ab77d9f`.

The retained Lean interface files identify the inherited logical calibration. They are reference sources, not a complete standalone Lean project; their full transitive imports were not rebuilt for this release.

References

[1] ProveIt research packet 127, *Shared arithmetic certificates and graph energies for Con(ZFC)*, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61/Computability/DiophantineConsistency/con-zfc-diophantine-forms/127-graph-certificates.

[2] ProveIt research packet 131, *Shared Pell caps and coprime product equations for ZFC consistency*, retained at the same pin. https://github.com/VladimirReshetnikov/ProveIt/tree/1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61/Computability/DiophantineConsistency/con-zfc-diophantine-forms/131-shared-pell-caps-round16.

[3] ProveIt, *Con(ZFC) in Diophantine form*, catalogue and active-research review, snapshot 1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61. https://github.com/VladimirReshetnikov/ProveIt/tree/1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61/Computability/DiophantineConsistency.

[4] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic 47 (1982), 549–571. https://doi.org/10.2307/2273588.

[5] J. Bayer, M. David, M. Hassler, Y. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/abs/2506.20909v3.